

Some aspects of sustained training of rainbow trout, *Salmo gairdneri* Richardson

R. NAHHAS, N. V. JONES AND G. GOLDSPIK

Zoology Department, University of Hull, Hull HU6 7RX, U.K.

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Groups of 6–7 cm length rainbow trout, *Salmo gairdneri* Richardson, were simultaneously trained at four water velocities (0, 1.4, 2.2 and 3.5 Ls^{-1}) for a period of 46 days. Oxygen consumption and swimming ability (fatigue time) were then measured. Only training at 3.5 Ls^{-1} increased the swimming ability of the fish. A study of the relative proportion of the white and red muscles indicated that the white muscle was increasing its mass at velocities in excess of 2.2 Ls^{-1} . The oxygen consumption rate of the trained fish was lower than that of the untrained fish when considered over the whole velocity range.

I. INTRODUCTION

Exercise training of fish is known to increase their ability to resist fatigue (Brett *et al.*, 1958; Hammond & Hickman, 1966), and to increase their critical swimming speed (Farlinger & Beamish, 1978).

Fish seem to achieve this by adaptive changes in their metabolic activity and/or their muscular morphology. Hochachka (1961) found that trained rainbow trout developed an oxygen debt which was three times higher than that developed by untrained fish when both groups were similarly exercised.

Greer-Walker (1971), Greer-Walker & Pull (1973), Greer-Walker & Emerson (1978) and Davison & Goldspink (1977, 1978) showed that prolonged training resulted in hypertrophy in the skeletal muscle of fish. The extent of hypertrophy and the change in the number of fibres were taken as measures of the 'involvement' of certain muscle types in activity at certain swimming speeds.

It is believed that the red fibres are involved in the aerobic slow swimming of fish (Bone, 1966; Johnston *et al.*, 1977), whereas the white fibres are involved in faster swimming particularly burst swimming. However, some evidence suggests that in teleosts the white fibres might be involved in speeds intermediate between slow and burst swimming (Hudson, 1973). Johnston & Moon (1980) have confirmed that the white muscle fibre of brook trout became active between 1.5 and 1.8 Ls^{-1} (body lengths per second). They have also shown that the red muscle fibres were active at all speeds $\leq 5 \text{ Ls}^{-1}$. This involvement of the white fibres may be associated with the polyneuronal innervation of the white fibres in teleosts.

The present paper is the first in a series of papers concerned with growth and training of rainbow trout under controlled conditions and, therefore, the training system used is described in some detail. The effect of training on swimming ability and the relationship of this to some adaptive changes are also discussed.

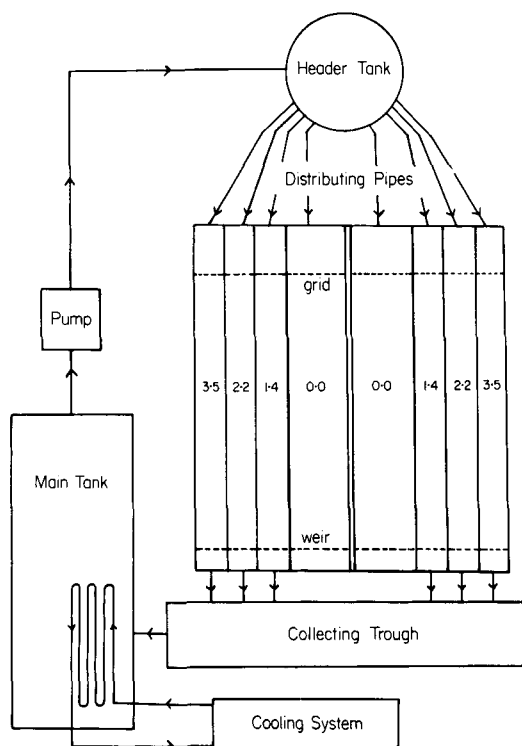


FIG. 1. A diagrammatic illustration of the training system. The numbers indicate the relative swimming velocities.

II. MATERIALS AND METHODS

THE TRAINING SYSTEM AND FISH TRAINING

A multi-channel exercise system was constructed. This system has the advantage of training small fish at four water velocities at the same time, and of duplicating each water velocity. The system is diagrammatically illustrated in Fig. 1. The system was balanced so that the header tank (plastic container: top diameter 42 cm, bottom diameter 38 cm, depth 58.5 cm) was always about one-third full of water. Twenty distributing pipes of the same cross section (1.5 cm) were connected around the circumference of the header tank at the same level (3 cm from the bottom). This arrangement allowed for equal flow of water in all the pipes which were distributed among the eight exercise channels. The channels were formed by placing rectangular pieces of marine plywood (122×15 cm) in parallel inside a shallow fibreglass tank ($122 \times 61 \times 15$ cm). The overflow from the exercise channels was collected in a trough ($90 \times 30 \times 30$ cm) which discharged the water back into the main tank ($91.5 \times 51.5 \times 38.0$ cm). Water was pumped from the main tank to the header tank by means of an in-line pump ('Mopump', type S28, Crane Ltd.).

The still water channels were closed from both ends but the walls of the exercise channels were not sealed to the bottom of the holding tank. This allowed water exchange among all channels and kept both temperature and dissolved oxygen uniformly distributed. It was anticipated that the fish kept in still water would require more room for spontaneous activity if they were not to be distressed. Therefore, the still water channels were made three times wider than the exercise channels—the latter being only 3.5 cm wide to allow for maximum uniformity of current flow as found by trial and error, and at the same time to accommodate the fish.

The water velocity was regulated by the number of pipes discharging water into each channel, and by the use of grids and weirs made of an adjustable material. The water velocity in cm s^{-1} was measured by means of a Kent miniflow current meter (Kent instruments (Stroud) Ltd.). The velocity was then expressed in fish body lengths per second (Ls^{-1}).

A closed refrigerated system with water at 6°C was available. From this system water was passed through a series of brass or glass tubes which were placed in the main tank of the training system to act as a cooling coil to control temperature. Oxygen was kept at 100% saturation by a supply of air to the system particularly to the sections of reduced water flow. The pH was regularly inspected and found to remain near 8.4.

One-fifth of the water in the system was replaced daily by fresh water taken directly from a mains water supply.

Rainbow trout, *Salmo gairdneri* Richardson *c.* 6 cm in length were trained at velocities of 0, 1.4, 2.2 and 3.5 Ls^{-1} . Five fish were used per channel and thus 10 fish were used at each water velocity. Each morning the fish were removed from the system, placed in tanks and given a ration of dry food to satiation. After feeding they were given a short rest when they were returned to the training system. The whole feeding operation took 1 h. Training lasted for 46 days after which the fish were subjected to a swimming test.

THE SWIMMING TEST

A flume described by Arnold (1969) was originally used by Johnston & Goldspink (1972) for exercise studies. The same flume was available for use in the present study with modification of the exercise trough. A narrower trough with a rounded bottom was fixed inside the main exercise trough. A water velocity of *c.* 70 cm s^{-1} combined with a uniform flow of water was achieved in the inner trough. This velocity was equivalent to *c.* 10 Ls^{-1} . This is higher than all the critical swimming speeds and also higher than all the burst swimming speeds listed by Beamish (1978) for rainbow trout, *S. gairdneri* Richardson.

The fish were tested by determining the fatigue time of a group of fish swimming at this high velocity. To do this, the fish were introduced to the swimming trough while the water velocity was kept below 0.5 Ls^{-1} by reducing the pumping rate and by means of weirs placed at the end of the main trough. The fish were left for 30 min before the water velocity was increased to equal that at which the fish were originally swimming. They were left at this velocity for a further 30 min. Then the maximum desired velocity of *c.* 10 Ls^{-1} was reached within 20 s of activating the main pumps and timing was started. The fatigue time of each fish was recorded and the average fatigue time of the group was calculated and expressed as seconds per gram of fish.

THE RELATIVE PROPORTION OF RED AND WHITE MUSCLE

To determine the proportion of red to white muscle, the technique of Broughton *et al.* (1981) was adopted with the following minor modification. Thirty cross sections from the area exactly behind the dorsal fin were cut and stained for lipids to differentiate the two muscle types. Each section was projected on to paper using a projection microscope. The sections were too large for the microscope's field of view and, therefore, only the top right-hand quarter of each section was used. Three fish were sampled from each water velocity for the above determinations.

OXYGEN CONSUMPTION IN RELATION TO TRAINING

Groups of six rainbow trout of an average length of 7 cm were trained at 0.5, 1.4, 2.2, and 3.5 Ls^{-1} for a period of one month. A control group of 30 fish was kept in still water for a similar period of time. All groups received food to satiation.

Oxygen consumption was determined using a Blazka (Blazka *et al.*, 1960) type respirometer as described by Broughton *et al.* (1980). Each group of the trained fish was introduced to the respirometer and left to acclimatize to the surroundings for 30 min during which the water velocity was kept to a minimum ($< 0.5 \text{ Ls}^{-1}$). Then the velocity was brought up to equal the original training velocity while the respirometer water was brought

to full air saturation. Timing was then started after stopping the air supply. After 20 min, the drop in oxygen was recorded. No change in oxygen occurred within a 20-min period when the system was run without fish present.

The untrained fish were divided into four groups and each was introduced to the respirometer to determine oxygen consumption at a velocity equivalent to one of the velocities of the training fish. The fish were introduced to the respirometer in a similar manner to the trained fish, but velocity increments of 0.5 Ls^{-1} were achieved every 10 min until the desired velocity was reached. Six determinations per group were made for both the trained and untrained fish. Each group was given a 3 h rest before the next determination.

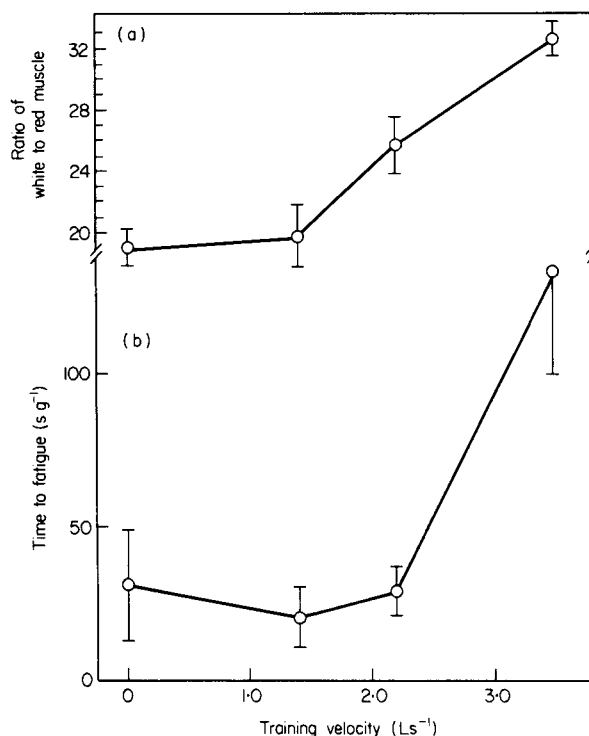


FIG. 2. The effect of training on the swimming ability of rainbow trout (a) The ratio of white to red muscle plotted against training velocity. (b) The time to fatigue at high swimming velocity plotted against training velocity. Bars indicate standard deviation.

III. RESULTS

Figure 2 shows the effect of the training velocity on the fatigue time at the high water velocity. Training velocities $< 2.5 \text{ Ls}^{-1}$ had no effect on the performance of the fish and the differences were not statistically significant ($P \gg 0.05$). At a training velocity of 3.5 Ls^{-1} the time to fatigue was increased by about 420% compared to the fish at the lower training velocities. The difference is very significant ($P \ll 0.001$).

The same figure shows the ratio of the white muscle to the red muscle plotted against training velocity. There was an increase in this ratio with increasing training velocity. Fish trained at 1.4 Ls^{-1} showed an increase of this ratio of about 4.7% compared to the fish in still water (at 0 Ls^{-1}). This increase was not significant ($P \gg 0.05$). Fish at 2.2 Ls^{-1} showed an increase of 35.6% and those at 3.5 Ls^{-1} showed an increase of 71.7% when compared to the fish kept in still water.

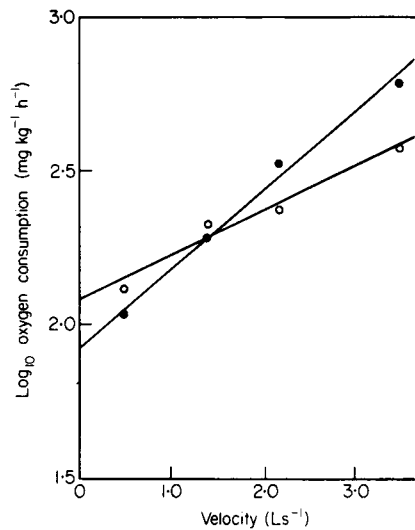


FIG. 3. Logarithm of oxygen consumption plotted against velocity for trained and untrained Rainbow trout. ○, Trained; ●, untrained. Regression equations: trained, $y = 2.08 + 0.15x$ ($r = 0.98$); untrained, $y = 1.92 + 0.26x$ ($r = 0.99$).

In both cases, the increase of the white to red muscle ratios was significant when compared to the fish in still water and to those at 1.4 Ls^{-1} ($P \ll 0.01$).

Figure 3 shows plots of the logarithm of oxygen consumption against water velocity for trained and untrained fish. A linear relationship with good correlation existed in both cases. By extrapolation to 0 velocity, the standard metabolic rate of the trained fish was found to be higher than that of the untrained fish. These figures were 120 and $83 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$, respectively (0.452 and 0.313 W kg^{-1}). The metabolic rate of the trained fish remained higher than that of the untrained fish up to a velocity of 1.43 Ls^{-1} when both metabolic rates were equal. Beyond 1.43 Ls^{-1} , the metabolic rate of the trained fish was lower than that of the untrained fish so that the level of metabolism at 3.5 Ls^{-1} for the untrained fish was almost 1.6 times that of the trained fish. This difference was very significant ($P < 0.01$). The difference between trained and untrained fish at 2.2 Ls^{-1} was less significant ($P < 0.05$), whereas at 1.4 and 0.5 Ls^{-1} there was no significant difference between the trained and the untrained fish ($P \gg 0.05$). There was a significant difference ($P < 0.01$) in the oxygen consumption of groups of fish swimming at different water velocities both in the case of trained and untrained fish.

IV. DISCUSSION

Training at 3.5 Ls^{-1} was shown to change the performance of the fish considerably (Fig. 2). It is likely that at such a relatively high speed, the white muscle fibres were recruited for swimming activity and thus they were effectively trained so as to enable the fish to show a better resistance to fatigue. The relative and significant increase in the white muscle of fish at this velocity when compared to those at lower velocities confirms that effective training of the white fibres was taking place at this velocity. Despite the significant change in the

proportion of the red and white muscles at 2.2 Ls^{-1} (Fig. 2) fish trained at this velocity were not as capable of swimming as those at 3.5 Ls^{-1} . It is possible that only when a certain number of fibres is recruited that effective training can take place. It may be that some white muscle fibres were partially recruited only beyond 1.4 Ls^{-1} , and only beyond 2.2 Ls^{-1} was full recruitment of some of the fibres taking place.

These findings support the works reviewed in the introduction to this paper, in that training increases the ability of the fish to resist fatigue when confronted with a drastic change in the water velocity. They also confirm the figures suggested in earlier works on rainbow trout. The finding by Hudson (1973) that 3.0 Ls^{-1} seem to be the speed above which the white fibres became active. Greer-Walker (1971) in his comment on one of Bainbridges's experiments anticipated that a swimming velocity above 2.0 Ls^{-1} was required to produce effective training. Later, Greer-Walker & Emerson (1978) found that maximum hypertrophy of trout white muscle occurred at 3.0 Ls^{-1} when compared to lower speeds. At the same speed the red muscle fibre diameter declined and this was attributed to the possibility of the contraction of these fibres above their optimal frequency.

Differences in oxygen consumption rates were only significant at velocities higher than 1.4 Ls^{-1} . The trained fish had a lower consumption rate than the untrained fish. It is possible that the trained fish were more capable of tolerating a higher oxygen debt than the untrained fish (Hochachka, 1961). Furthermore, a change in the metabolism of the fish as a result of the training regime is possible. For example, fish sampled from running water were shown to accumulate more glycogen than those sampled from still water (Broughton & Goldspink, 1978) indicating more energy from anaerobic pathways.

Below 1.4 Ls^{-1} differences in oxygen consumption rates were not significant, however, the trained fish here had a higher consumption rate. The standard metabolic rate of the trained fish was also higher. This agrees with Washbourn (cited in Hochachka, 1961). It is very likely that up to 1.4 Ls^{-1} oxidative metabolism was predominant in the red muscle. Training, therefore, would only enhance the functional properties of this muscle at these low velocities. Beyond 1.4 Ls^{-1} , the white muscle seem to take an increasing role in metabolism. Johnston & Moon (1980) found that the threshold speed for recruiting white fibres in brook trout, *Salvelinus fontinalis*, was between 1.5 and 1.8 Ls^{-1} . The white muscle is basically anaerobic and, therefore, training of this muscle might emphasize this anaerobic nature. The total oxygen consumption is, therefore, reduced possibly because some balance between the red and white muscle demands exist beyond 1.4 Ls^{-1} . Again, Johnston & Moon (1980) found that both these muscles were functional at speeds $> 2.0 \text{ Ls}^{-1}$. They also found a marked hypertrophy of the red muscle at 3.0 Ls^{-1} .

In experiments examining the effect of training on trout growth (Nahhas, 1981; Nahhas *et al.*, in prep.) it was found that both quantitative and qualitative changes in the specific growth rates did take place depending on the relative training velocities. In particular, it was found that changes were most pronounced between 2.2 and 3.5 Ls^{-1} . At the latter velocity growth was achieved by an increase in length in preference to weight, whereas at lower velocities growth was either achieved by increasing both weight and length or emphasis was laid on weight increase alone. It is likely that at 3.5 Ls^{-1} , the increase in the white

muscle mass was achieved by increasing the number of fibres rather than (or in addition to) their diameters. There is evidence that this increase in white muscle mass was accompanied by a high protein content of the dry muscle (Nahhas, 1981). It was also reported by Weatherley *et al.* (1979) that the growth of mosaic muscle of yearling rainbow trout was mostly achieved by an increase in fibre number. More structural and biochemical studies will be necessary before the exact nature of the training effect is known.

References

- Arnold, G. P. (1969). A flume for behaviour studies of marine fish. *J. exp. Biol.* **51**, 671–679.
- Beamish, F. W. H. (1978). Swimming capacity. In *Fish Physiology* (W. S. Hoar & D. J. Randall), Vol. VII, pp. 101–187. Academic Press: London.
- Blazka, P., Volf, M. & Capela, M. (1960). A new type of respirometer for the determination of the metabolism of fish in an active state. *Physiologia Bohemoslov.* **9**, 553–560.
- Bone, Q. (1966). On the function of the two types of myotomal muscle fibre in elasmobranch fish. *J. Mar. Biol. Ass. U.K.* **46**, 321–349.
- Brett, J. R., Hollands, M. & Alderdice, D. F. (1958). The effect of temperature on the cruising speed of young sockeye and coho salmon. *J. Fish Res. Bd Can.* **15**, 587–605.
- Broughton, N. M. & Goldspink, G. (1978). Biochemical changes in the lateral muscles of roach *Rutilus rutilus* from two habitats following exercise. *J. Fish Biol.* **13**, 613–618.
- Broughton, N. M., Goldspink, G. & Jones, N. V. (1980). The effect of training on the lateral musculature of O-group roach, *Rutilus rutilus* (L), and their fatigue in subsequent exercise tests. *J. Fish Biol.* **17**, 209–217.
- Broughton, N. M., Goldspink, G. & Jones, N. V. (1981). Histological differences in the lateral musculature of O-group roach, *Rutilus rutilus* (L.) from different habitats. *J. Fish Biol.* **18**, 117–122.
- Davison, W. & Goldspink, G. (1977). The effect of prolonged exercise on the lateral musculature of the brown trout (*Salmo trutta*). *J. exp. Biol.* **70**, 1–12.
- Davison, W. & Goldspink, G. (1978). The effect of training on the swimming muscles of the goldfish (*Carassius auratus*). *J. exp. Biol.* **74**, 115–122.
- Farlinger, S. & Beamish, F. W. H. (1978). Changes in blood chemistry and critical swimming speed of largemouth bass, *Micropterus salmoides*, with physical conditioning. *Trans. Am. Fish. Soc.* **107**, 523–527.
- Greer-Walker, M. (1971). Effect of starvation and exercise on the skeletal muscle fibres of the cod (*Gadus morhua* L.) and the coalfish (*Gadus virens*) respectively. *J. Cons. Int. Explor. Med.* **33**, 421–426.
- Greer-Walker, M. & Emerson, L. (1978). Sustained swimming speeds and myotomal muscle function in the trout, *Salmo gairdneri*. *J. Fish Biol.* **13**, 475–481.
- Greer-Walker, M. & Pull, G. (1973). Skeletal muscle function and sustained swimming speeds in the coalfish *Gadus virens* L. *Comp. Biochem. Physiol.* **44A**, 495–501.
- Hammond, B. R. & Hickman, C. P. Jr. (1966). The effect of physical conditioning on the metabolism of lactate, phosphate, and glucose in rainbow trout, *Salmo gairdneri*. *J. Fish. Res. Bd Can.* **23**, 65–83.
- Hochachka, P. W. (1961). The effect of physical training on oxygen debt and glycogen reserves in trout. *Can. J. Zool.* **39**, 367–776.
- Hudson, R. C. L. (1973). On the function of the white muscles in the teleosts at intermediate swimming speeds. *J. Exp. Biol.* **58**, 509–522.
- Johnston, I. A., Davison, W. & Goldspink, G. (1977). Energy metabolism of carp swimming muscles. *J. Comp. Physiol.* **114B**, 203–216.

- Johnston, I. A. & Goldspink, G. (1972). The effect of sustained swimming on the concentration of glycogen and lactate occurring in the myotomal muscles and liver of the coalfish (*Gadus virens* L.) *J. Mar. Biol. Ass. U.K.* **53**, 17–26.
- Johnston, I. A. and Moon, T. W. (1980). Exercise training in skeletal muscle of brook trout (*Salvelinus fontinalis*). *J. exp. Biol.* **87**, 177–194.
- Nahhas, R. (1981). Studies of growth and swimming capabilities of young trout under controlled conditions. Ph.D. thesis, University of Hull, England.
- Weatherley, A. H., Gill, H. S. & Rogers, S. C. (1979). Growth dynamics of muscle fibres, dry weight and condition in relation to somatic growth rate in yearling rainbow trout (*Salmo gairdneri*). *Can. J. Zool.* **57**, 2385–2392.